



SVP-19-019

10 CFR 50.73

April 5, 2019

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D.C. 20555

Quad Cities Nuclear Power Station, Unit 1
Renewed Facility Operating License No. DPR-29
NRC Docket No. 50-254

Subject: Licensee Event Report 254/2019-001-00 "HPCI System Inoperable Due to Blown Fuse in the Motor Speed Changer Circuit"

Enclosed is Licensee Event Report 254/2019-001-00 "HPCI System Inoperable Due to Blown Fuse in the Motor Speed Changer Circuit", for Quad Cities Nuclear Power Station, Unit 1.

This report is submitted in accordance with 10 CFR 50.73(a)(2)(v)(D) for an event or condition that could have prevented fulfillment of a safety function to mitigate the consequences of an accident.

There are no regulatory commitments contained in this letter.

Should you have any questions concerning this report, please contact Mark Humphrey at (309) 227-2800.

Respectfully,

A handwritten signature in black ink, appearing to read "K. Ohr", written over a horizontal line.

Kenneth S. Ohr
Site Vice President
Quad Cities Nuclear Power Station

cc: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – Quad Cities Nuclear Power Station

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollect.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name

Quad Cities Nuclear Power Station Unit 1

2. Docket Number

05000254

3. Page

1 OF 4

4. Title

HPCI System Inoperable Due to Blown Fuse in the Motor Speed Changer Circuit

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name n/a	Docket Number
02	05	2019	2019	001	00	04	05	2019	Facility Name n/a	Docket Number 05000
9. Operating Mode			11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)							
1			☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)				
			☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)				
			☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)				
			☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)				
10. Power Level			☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)				
100			☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)				
			☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)				
			☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)				
			☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)				
			☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)						

12. Licensee Contact for this LER**Licensee Contact**

Richard Swart – Regulatory Assurance

Telephone Number (Include Area Code)

309-227-2810

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
X	BJ	33	G080	Y	X	BJ	MO	G080	Y

14. Supplemental Report Expected**15. Expected Submission Date**☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No

Month	Day	Year
n/a		

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On February 5, 2019, at 1804 hours, a Motor Speed Changer (MSC) logic fuse blew during performance of Quad Cities Nuclear Power Station (QCNPS) Unit 1 High Pressure Coolant Injection (HPCI) operability surveillance following planned maintenance activities. Had HPCI been demanded while the fuse was blown the HPCI turbine would have been unable to reach its required speed.

During investigation of the blown fuse, the MSC motor and the MSC gearbox were discovered to be mechanically bound. Intermittent failure of the Low Speed Stop (LSS) Limit Switch (LS) due to age related degradation of internal electrical contacts allowed the MSC motor to remain energized when it should have been deenergized, leading to degradation. The limit switch contacts were found with high resistance attributed to observed deposits and pitting. None of these components were part of the work scope of this HPCI maintenance window.

The immediate corrective action was replacement of the fuse, the LSS LS, the MSC motor and select MSC gearbox components. The safety impact was minimal based on the availability of other high pressure and low pressure Emergency Core Cooling Systems.

This is being reported in accordance with 10 CFR 50.73(a)(2)(v)(D) for an Event or Condition that Could Have Prevented Fulfillment of a Safety Function to Mitigate the Consequences of an Accident.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Quad Cities Nuclear Power Station Unit 1	05000254	2019	- 001	- 00

NARRATIVE**PLANT AND SYSTEM IDENTIFICATION**

General Electric - Boiling Water Reactor, 2957 Megawatts Thermal Rated Core Power

Energy Industry Identification System (EIIIS) codes are identified in the text as [XX].

EVENT IDENTIFICATION

HPCI System Inoperable Due to Blown Fuse in the Motor Speed Changer Circuit

A. CONDITION PRIOR TO EVENT

Unit: 1 Event Date: February 5, 2019 Event Time: 1804 hours
Reactor Mode: 1 Mode Name: Power Operation Power Level: 100%

There were no other structures, systems or components (SSC) inoperable during this event time period that could have contributed to this event.

B. DESCRIPTION OF EVENT

On February 5, 2019, at 1804 hours, during a Unit 1 High Pressure Coolant Injection (HPCI)[BJ] operability surveillance, a fuse [FU] blew in the logic for the Motor Speed Changer (MSC)[SM] for the HPCI Turbine [TRB]. The blown fuse disabled both the manual and automatic injection functions of HPCI. Had HPCI been demanded while the fuse was blown the HPCI turbine would have been unable to reach its required speed. The system had been out of service for planned maintenance, but no maintenance activities took place on equipment protected by this fuse. On February 6, 2019, at 0222 hours, ENS #53816 was made to the NRC under 10 CFR 50.72(b)(3)(v)(D), to report this event as an event or condition that could have prevented the fulfillment of a safety function. November 8, 2018 was the last successful test of Unit 1 HPCI.

During troubleshooting, it was found that the MSC motor [MO] and the MSC gearbox were mechanically bound. The fuse, the MSC motor, and select gearbox components were replaced. Subsequent troubleshooting determined that intermittent failure of the Low Speed Stop (LSS) Limit Switch (LS)[33], caused intermittent operation of the associated relay. The LS intermittent failure was due to high resistance contacts attributed to deposits and pitting caused by age related degradation. The LSS LS was replaced.

After troubleshooting and repairs, a HPCI operability surveillance was performed successfully. HPCI was declared operable on February 8, 2019, at 1434 hours. Since HPCI is a single train safety system, this notification is being made in accordance with 10 CFR 50.73(a)(2)(v)(D) for an Event or Condition that Could Have Prevented Fulfillment of a Safety Function to Mitigate the Consequences of an Accident.

C. CAUSE OF EVENT

The cause of the event is the intermittent failure of the MSC LSS LS, which impacted both the MSC motor and gearbox. LS failure is attributed to age related degradation of internal LS electrical contacts. The LS had been installed for more than 9 years and was due for replacement at 10 years.

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NARRATIVE**D. SAFETY ANALYSIS****System Design**

According to the Quad Cities Nuclear Power Station (QCNPS) Units 1 and 2 UFSAR Section 6.3.2.3, "High Pressure Coolant Injection Subsystem," the HPCI subsystem is designed to pump water into the reactor vessel under Loss of Coolant Accident (LOCA) conditions which do not result in rapid depressurization of the pressure vessel. The loss of coolant might be due to a loss of reactor feedwater or to a small line break which does not cause immediate depressurization of the reactor vessel. The sizing of the HPCI subsystem is based upon providing adequate core cooling during the time that the pressure in the reactor vessel decreases to a value that the Core Spray [BM] subsystem and/or the Low Pressure Coolant Injection (LPCI) [BO] subsystem become effective. The HPCI subsystem is designed to pump 5600 gallons per minute into the reactor vessel within a reactor pressure range of about 1120 pounds per square inch gage (psig) to 150 psig. Initiation of the HPCI subsystem occurs automatically on signals indicating reactor low-low water level or high drywell pressure. HPCI injection into the reactor vessel may be accomplished manually by the operator or without operator action by the HPCI automatic initiation circuitry. HPCI can also operate in a pressure control mode of consuming steam from the reactor vessel without providing full injection into the vessel (down to and including zero injection).

Safety Impact

The safety significance of this event is minimal. HPCI was inoperable for approximately 68.5 hours. The Reactor Core Isolation Cooling (RCIC)[BN] system, Automatic Depressurization System (ADS), and multiple low-pressure Emergency Core Cooling Systems (ECCS) were available during this time.

Risk Insights

The HPCI system is a risk significant system in the QCNPS PRA, supports High Safety Significant and Low Safety Significant Maintenance Rule Functions and is monitored under the Mitigating System Performance Index (MSPI) for reliability and availability. Unavailability of the HPCI system during this condition was addressed under the QCNPS Configuration Risk Management program per the requirements of 10CFR50.65 (a)(4) to assess and manage the risk increase.

This event is a Maintenance Rule Functional Failure (MRFF), and a MSPI failure, since both the manual and automatic injection functions of HPCI were lost.

E. CORRECTIVE ACTIONS**Immediate:**

1. Replaced the LSS LS, MSC Motor, and select MSC gearbox components.
2. Failed LS and Motor were sent to Exelon PowerLabs for further failure analysis.

Follow-up:

1. The preventive maintenance replacement frequency of the LSS LS will be reduced from 10 years to 8 years.

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NARRATIVE**F. PREVIOUS OCCURRENCES**

The station events database, LERs, and INPO Consolidated Event System (ICES) were reviewed for similar events at QCNPS. This event was attributed to intermittent LSS LS failure resulting in MSC motor and gearbox binding and a blown fuse.

Quad Cities Unit 1 LER 05000254/95-004-00, "HPCI System Declared Inoperable From Motor Speed Changer Malfunction Due to Limit Switch Failure," involved the same system components as the current event. In 1995, the LSS LS was found to have improper setting of the internal contact block, which also resulted in intermittent performance, causing MSC damage. Corrective actions addressed proper maintenance instructions for setting of the limit switches. The corrective actions from the 1995 event were not found to contribute to the current event.

Quad Cities Unit 2 Issue Report 593863, "MSC Full Reposition Greater Than 15 Seconds," also involved the same system components as the current event. In 2007, a loose screw associated with the LSS LS allowed the MSC motor to remain energized and driving into the LSS since a logic test six days earlier. Unlike the current event, it did not result in damage to the MSC motor or gear box, and HPCI remained capable of meeting its design functions. Corrective actions addressed verifying that the MSC motor has de-energized after returning HPCI to standby. The corrective actions from the 2007 event were not found to contribute to the current event.

G. COMPONENT FAILURE DATA

Failed Equipment: Limit Switch, Series A
Component Manufacturer: General Electric
Component Model Number: CR9440K1K1
Component Part Number: N/A

Failed Equipment: MSC Motor, 1/10 HP, 125 VDC
Component Manufacturer: General Electric
Component Model Number: 5BC26AC389B
Component Part Number: N/A

This event has been reported to ICES.